TrueMark

Secure Digital Truth in a Synthetic World

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The Problem

Why does digital image protection matter today?

Watermark Removal

Visible watermarks are trivially stripped via cropping or AI inpainting tools

Metadata Tampering

EXIF and metadata can be deleted or forged using freely available software

AI Content Scraping

Creators' images used in AI training datasets without consent or compensation

Deepfake Crisis

Advanced manipulation tools make it impossible to trust visual content

What is TrueMark?

TrueMark is a cross-platform mobile security suite that embeds protection directly into digital images using cryptographic techniques and invisible watermarking.

🔏 Ownership Protection

Verifiable proof of authorship for creators

🔒 Privacy & Security

Encryption + integrity for sensitive content

🤖 Ethical AI Usage

Machine-readable signals to prevent AI scraping

5

Security Modules

AES-256

Military-Grade Encryption

SHA-256

Content Fingerprinting

Core Modules

Five specialized security tools in a unified app

TrueSign

Embeds invisible cryptographic ownership signatures into images using LSB steganography. Links each file to its creator via a Firestore registry.

TrueMeta

Analyzes and sanitizes image metadata — strips EXIF, GPS coordinates, and device identifiers to prevent privacy leakage.

TrueLock

Encrypts any file into a secure .tmk container using AES-256-CBC with PKCS7 padding and a password-derived key via PBKDF2.

TrueVault

PIN-protected local secure storage for all generated artifacts, with filtering, sorting, and export capabilities.

TrueHide

Conceals secret text or binary data inside images using LSB pixel steganography with AES-256 payload encryption.

System Architecture

Frontend (Flutter)

Material Design 3 UI
Flutter SDK 3.8.1+
TrueSign · TrueLock · TrueHide
TrueMeta · TrueVault screens

Cryptographic Layer

AES-256-CBC/GCM Encryption
SHA-256 Hashing
LSB Steganography
PBKDF2 Key Derivation

Firebase Backend

Firebase Authentication
Cloud Firestore Registry
REST API fallback
for Windows

End-to-End Security Flow

1

File Selection

User captures or imports image from device or TrueVault

2

Metadata Scan

Sensitive EXIF, GPS, device info detected and optionally sanitized

3

Encryption

AES-256-GCM encrypts file; PBKDF2 derives key from user password

4

Hash & Embed

SHA-256 fingerprint generated; signature embedded via LSB steganography

5

Registry Storage

Hash + ownership info stored in Cloud Firestore for future verification

6

Secure Output

Protected file saved to TrueVault or exported to device

7

Verification

Embedded data extracted, hash recomputed and matched against registry

Technology Stack

Frontend

- Flutter SDK 3.8.1+ · Dart 3.8.1
- Material Design 3 (Indigo theme)
- image_picker · flutter_image_compress
- crypto · encrypt · pointycastle libs

Cryptography

- AES-256-CBC / AES-256-GCM
- SHA-256 content fingerprinting
- PBKDF2-SHA256 key derivation
- CRC32 integrity checksums

Backend

- Firebase Authentication
- Cloud Firestore (NoSQL)
- REST API fallback (Windows)
- OS-level secure storage (PIN)

Security Techniques

- LSB (Least Significant Bit) steganography
- Structured payload with length + CRC headers
- Random IV per encryption operation
- Zero-trust client-side processing model

Results & Comparative Analysis

Feature	Traditional Tools	TrueMark
Encryption	Basic AES (no auth)	AES-256-GCM (confidentiality + integrity)
Data Hiding	Rare / separate tools	Integrated LSB steganography
Integrity Check	Limited	SHA-256 verification
Metadata Handling	Often ignored	Deep analysis & sanitization
Ownership Verification	Not available	Firestore-based registry
Secure Storage	Basic file storage	TrueVault (sandboxed + PIN)

Limitations & Future Roadmap

Current Limitations

⚠ LSB steganography fails under JPEG compression, resizing, or format conversion

⚠ Centralized Firestore registry creates single point of trust dependency

⚠ Password-based security (no biometrics yet); no batch processing support

⚠ No asymmetric digital signatures — limits third-party verification

Future Enhancements

→ DCT/DWT frequency-domain watermarking for compression robustness

→ Blockchain-based decentralized ownership registry

→ AI/CNN-based deepfake and image forgery detection

→ ECC digital signatures + biometric authentication

Conclusion

TrueMark demonstrates that complex, multi-layered digital security can be made accessible to everyday users through thoughtful design and modern cryptography.

Integrated Pipeline

Sign → Lock → Hide → Analyze → Store in one app

Client-Side Privacy

Sensitive ops stay on device; only hashes reach cloud

Accessible Security

Complex cryptography behind an intuitive mobile UI

Built on: Flutter · Firebase · AES-256 · SHA-256 · LSB Steganography · Firestore

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